	stores the same amount of charge, capacitor Y has 2 μC
20	**A** Time constant = RC = 4700 x 2200 x 10^{-6} Since energy stored in capacitor, $E = \frac{1}{2}CV^2$, E is proportional for V^2 $\frac{E_o}{E_t} = \left(\frac{V_o}{V_t}\right)^2 = 8$ $\frac{V_o}{V_t} = \sqrt{8}$ Apply the equation for a discharging capacitor $V_t = V_o e^{-t/\tau}$ $\frac{1}{\sqrt{8}} = e^{-t/(4700\ x\ 2200\ x\ 10^{-6})}$ Solving, t = 10.8 s ≈ 11 s
21	**B**